Walk-In Pantry Layout Study

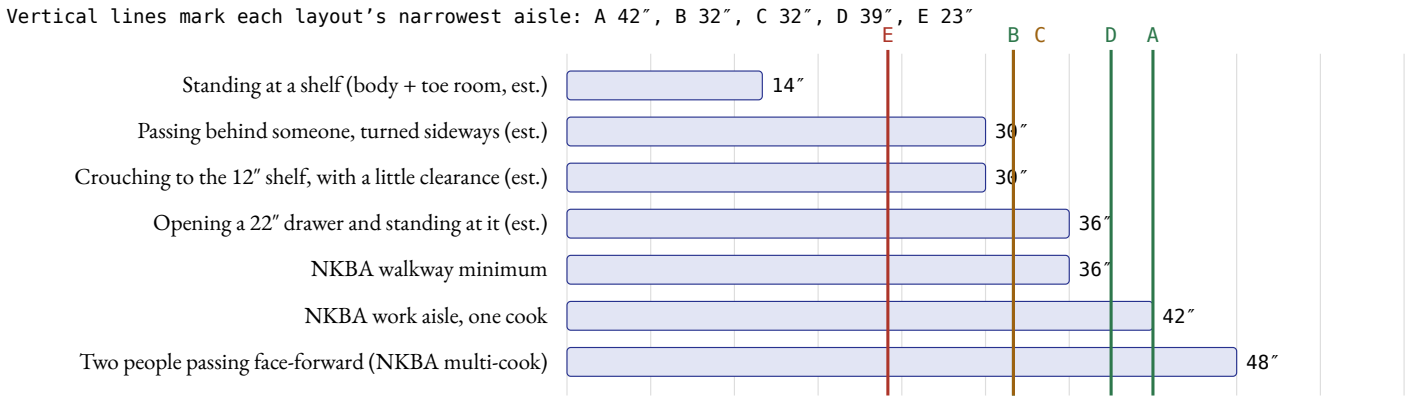
13′6″ × 6′ Pantry

Five layouts for the 162″ × 72″ pantry with an 8′ ceiling and a 32″ French door that swings into the kitchen. Shelves are 15″ deep at 12″, 27″, 42″, 57″, and 72″. The workspace is 25″ deep at 36″ and sized for a Bosch Universal Plus, a blender, and a mixer. Every drawing and rendering comes from one to-scale model.

Recommendation

Build Layout A. It is the only layout that keeps a 42″ aisle everywhere, which is NKBA's work-aisle guideline for one cook, and it puts the baker at the dead end of the room instead of in the path to the shelves. The counter goes on the wall where the old 28″ opening is being closed, so outlets and shelf blocking go into new framing. Layout B is the runner-up if a short carry from the kitchen matters more than a pinch-free aisle; it gains about 10 sq ft of shelf but pinches to 32″ at the counter ends.

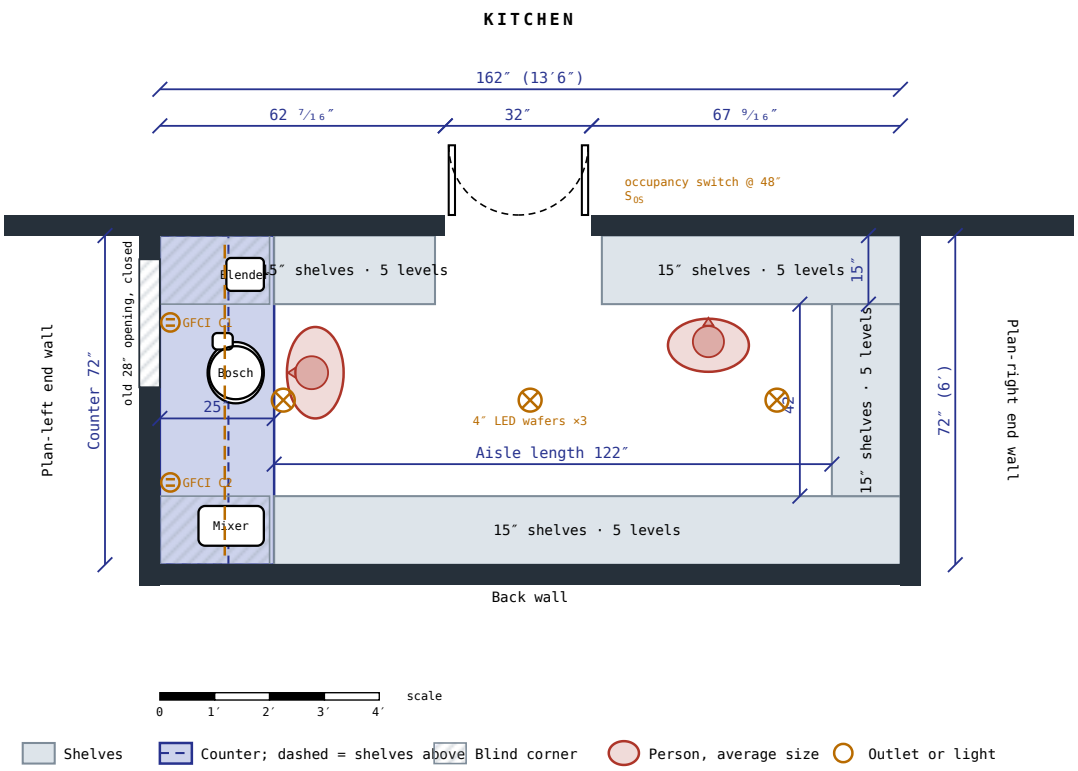
Layout	*Verdict*	*Narrowest Aisle*	*Counter*	*Shelf Surface*	*Drawer Bases*	*Best For*
A End-Wall Workstation	● Recommended	42″	72″	161 sq ft	42″	Baking station plus lots of storage
B Counter Facing the Door	● Realistic	32″	72″	171 sq ft	72″	Quick carry to and from the kitchen
C Full-Length Back Counter	● Tight	32″	162″	132 sq ft	132″	Most counter, one person at a time
D Butler Counter + Can Wall	● Realistic	39″	162″	109 sq ft	132″	Counter-heavy use; cans and jars one deep
E Deep Everything	● Not realistic	23″	162″	177 sq ft	114″	Do not build



Layout A: End-Wall Workstation

● Recommended

The counter takes the plan-left end wall, where the old opening gets closed. Shelves line the other three walls, and the aisle stays 42″ wide everywhere.



Narrowest aisle	**42″**	*Counter*	**72″**
Shelf surface	**161 sq ft**	*Running shelf*	**128 ft**
Drawer bases	**42″**	*Blind corner*	**30″**

What Works

- 42″ aisle from end to end: room to crouch at the bottom shelf, and a second person can slip past sideways.
- Whoever is baking stands at the dead end, out of the path to the shelves.
- The counter wall is the one being closed up, so outlets and shelf blocking go into new framing instead of finished drywall.
- 72″ of counter fits the Bosch, blender, and mixer with about 22″ of open work space between them.

What to Watch

- The counter is a 5-foot walk from the door.
- About 10 sq ft less shelf surface than Layout B.
- The first and last 15″ under the counter sit behind the side shelves. Those two blind corners suit rarely used items.
- Two inside corners on the plan-right side are a long reach.



From the doorway



At the workstation

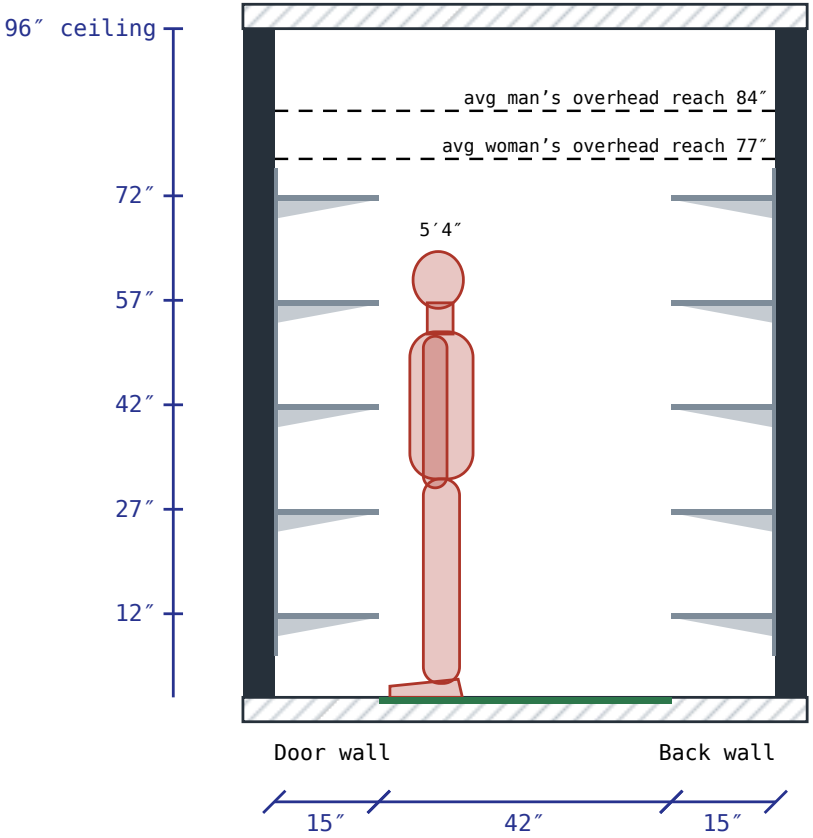


Overhead cutaway, kitchen at top

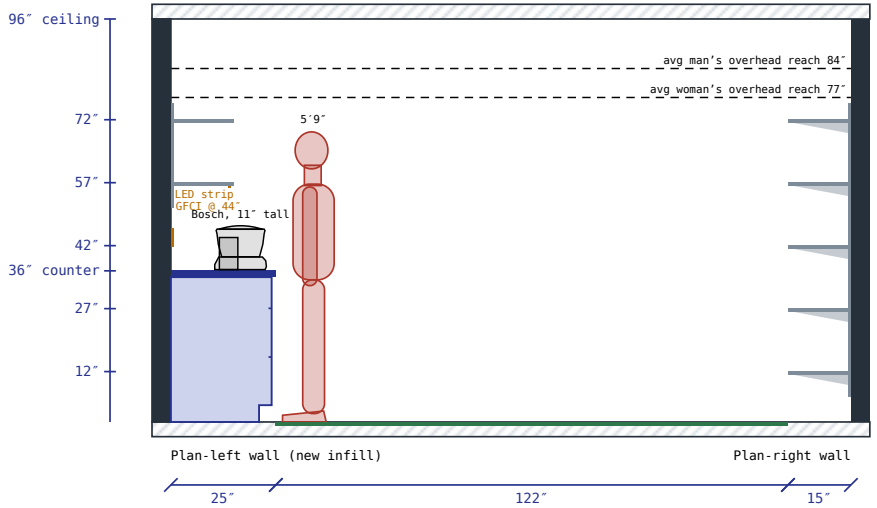
Layout A: Sections

Cut through the room and viewed from the side. Heights are inches above the floor; the colored floor bar marks the clear aisle.

Cross-Section Through the Aisle



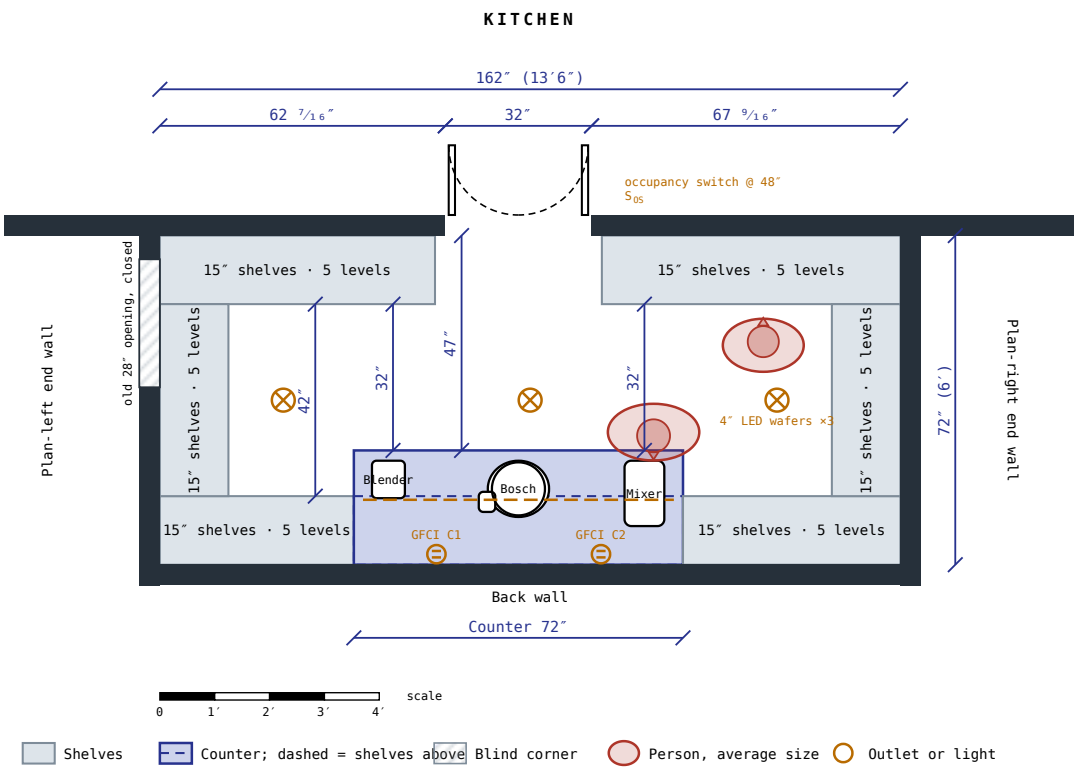
Long Section Through the Workstation



Layout B: Counter Facing the Door

Realistic

A 72″ counter sits on the back wall straight across from the door. Shelves wrap the rest of the room.



Narrowest aisle	**32″**	*Counter*	**72″**
Shelf surface	**171 sq ft**	*Running shelf*	**137 ft**
Drawer bases	**72″**	*Blind corner*	**none**

What Works

- Shortest carry: the counter is 47″ straight in from the door.
- You work in the open zone in front of the doorway, which is 47″ deep.
- Shelves on all four walls, with a 42″ aisle at the shelves.

What to Watch

- The aisle pinches to 32″ at each end of the counter, where the door-wall shelves face it.
- Whoever is at the counter stands in the path into the room.
- The appliances use most of the counter; about 16″ of open work space sits on each side of the Bosch.



From the doorway



At the workstation

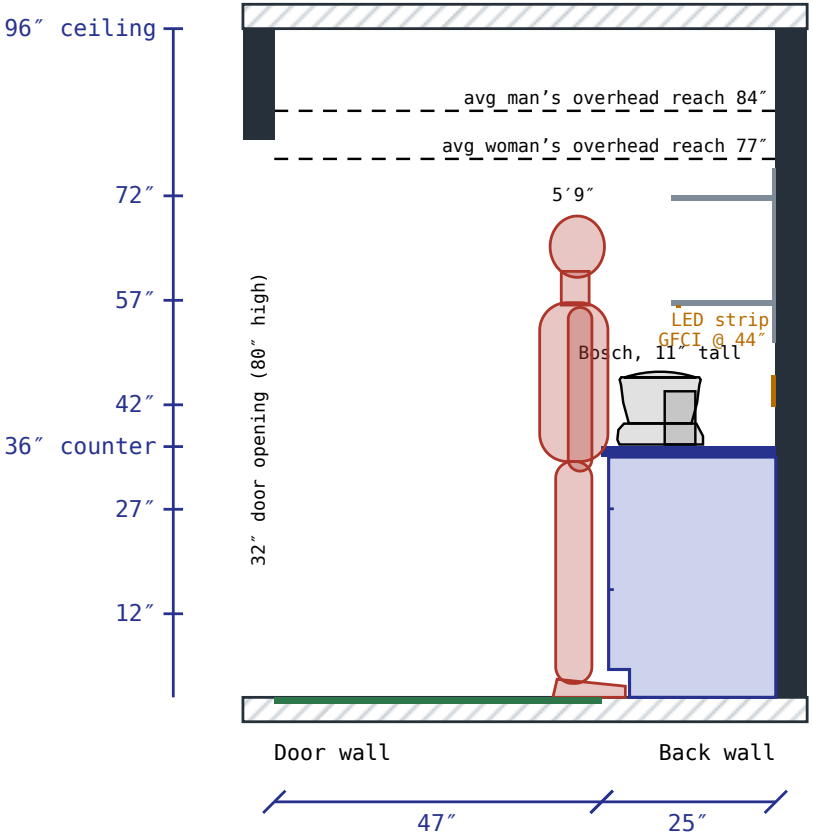


Overhead cutaway, kitchen at top

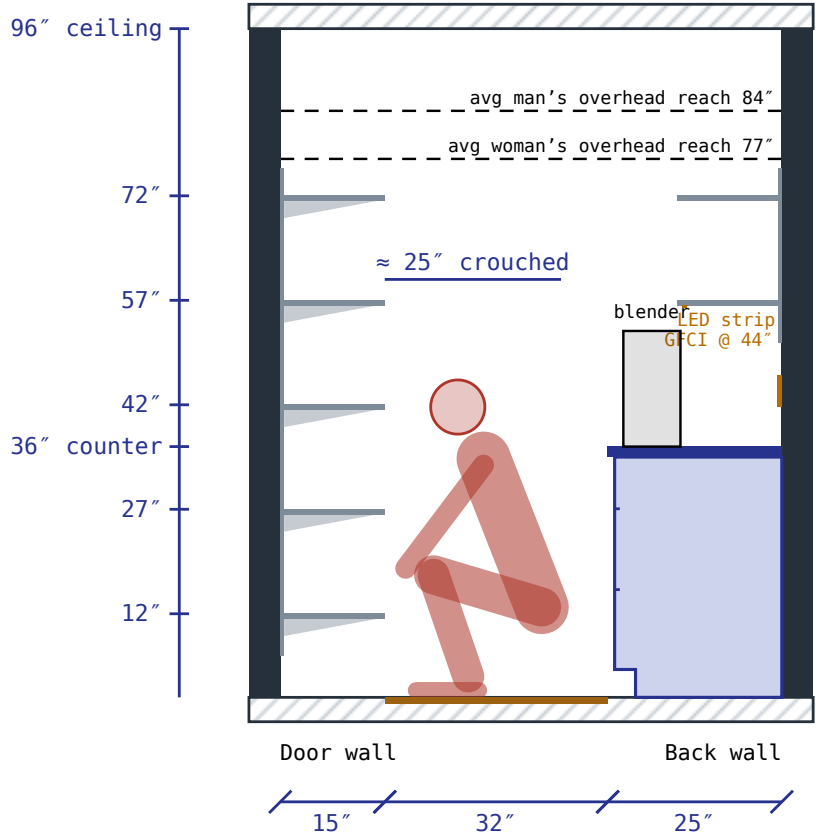
Layout B: Sections

Cut through the room and viewed from the side. Heights are inches above the floor; the colored floor bar marks the clear aisle.

Section Through the Door and Counter



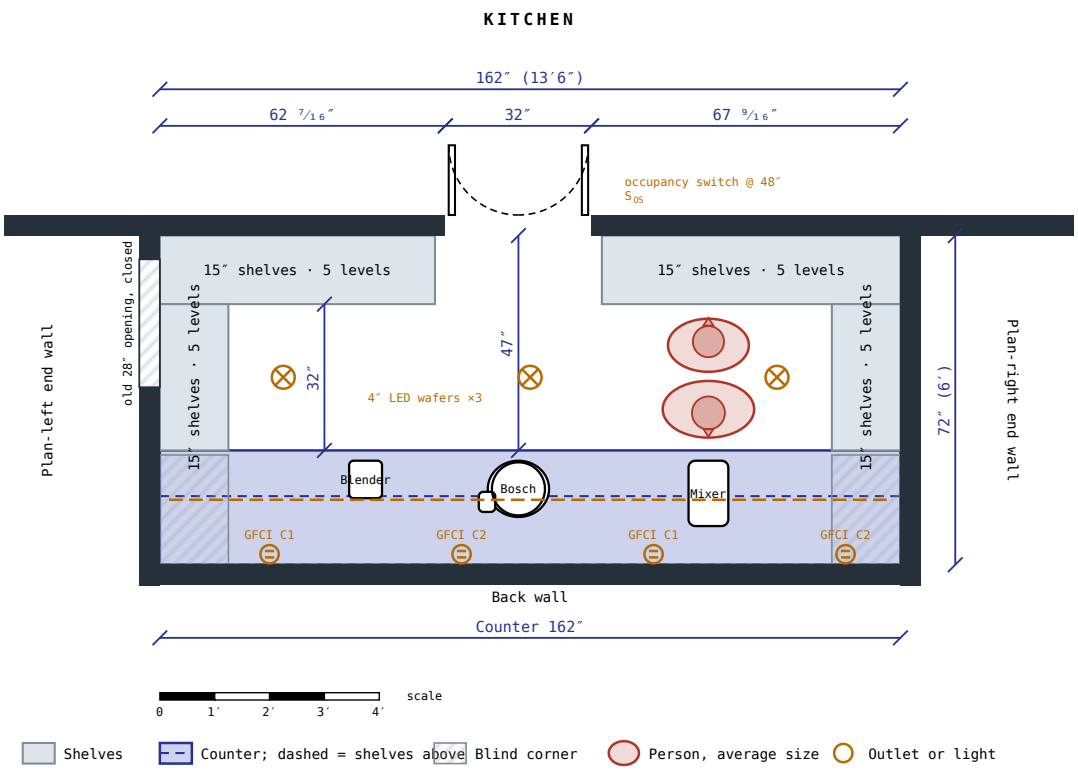
Section at the End of the Counter (the 32″ Pinch)



Layout C: Full-Length Back Counter

Tight

A 13½-foot counter with drawers runs the whole back wall. Shelves cover the door wall and the two ends.



Narrowest aisle	**32″**	*Counter*	**162″**
Shelf surface	**132 sq ft**	*Running shelf*	**106 ft**
Drawer bases	**132″**	*Blind corner*	**30″**

What Works

- 162″ of counter and 132″ of deep drawers.
- Room to leave every appliance out and still roll dough.

What to Watch

- 32″ aisle for the full length, under NKBA's 36″ walkway minimum.
- A 22″ drawer pulled open leaves 10″ to stand in, so you work from the side.
- Nobody passes anybody. Crouching at a door-wall shelf puts your back against the counter.



From the doorway



At the workstation

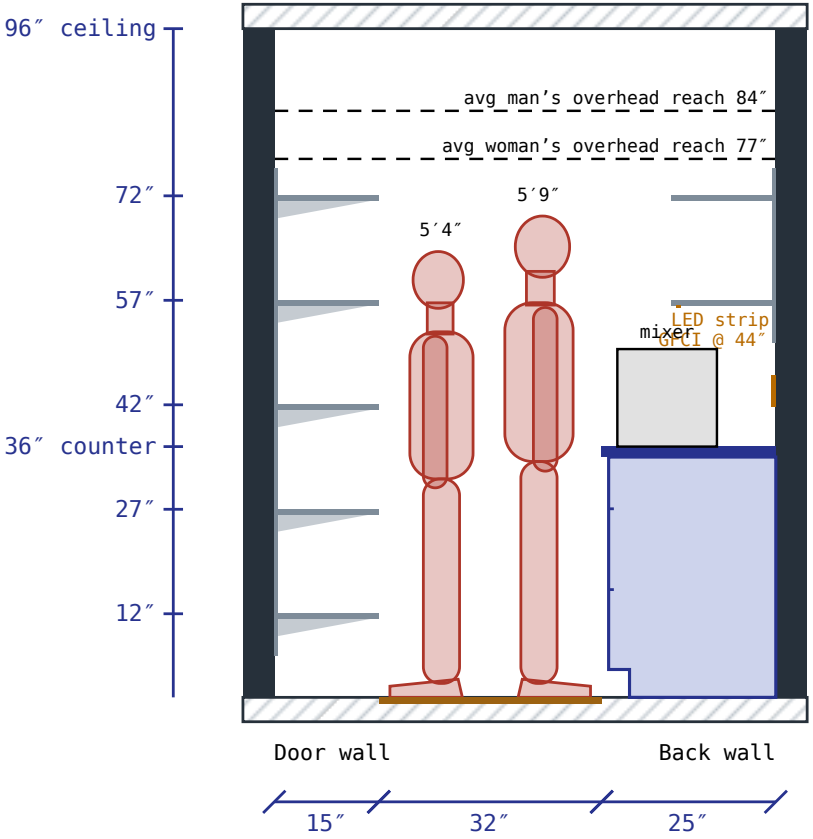


Overhead cutaway, kitchen at top

Layout C: Sections

Cut through the room and viewed from the side. Heights are inches above the floor; the colored floor bar marks the clear aisle.

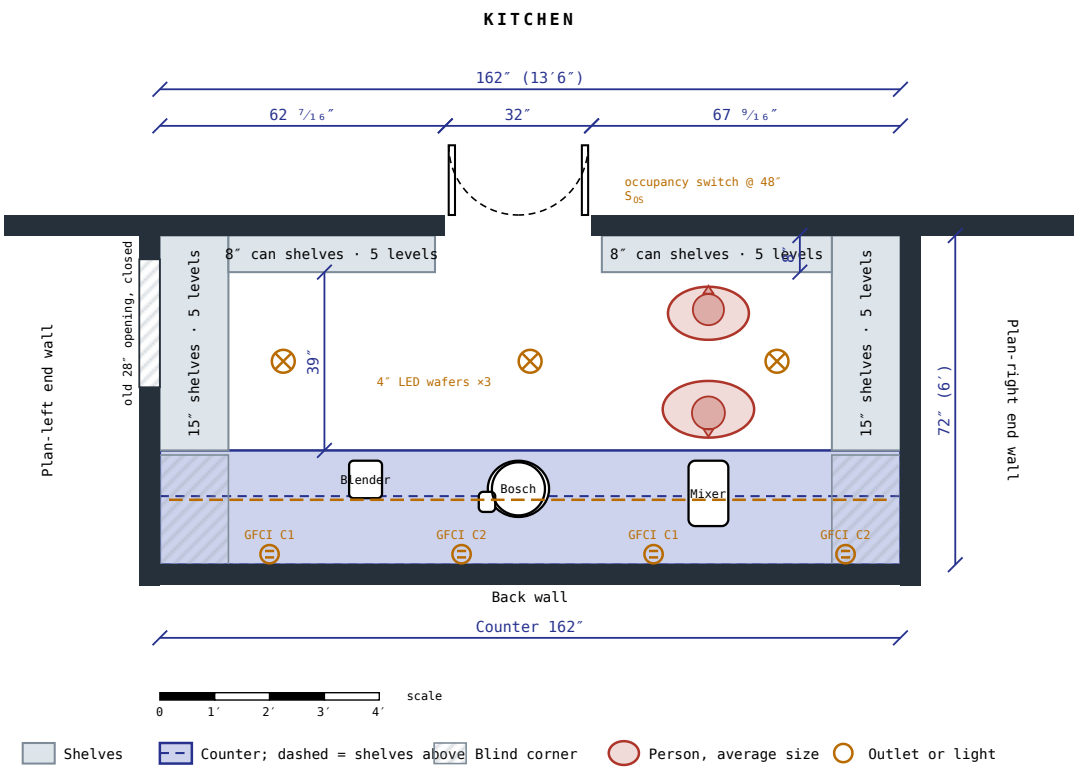
Cross-Section: Two People in a 32″ Aisle



Layout D: Butler Counter + Can Wall

Realistic

The full-length counter stays, but the door wall gets 8″ can-and-jar shelves instead of 15″ ones, which buys back a 39″ aisle.



Narrowest aisle	**39″**	*Counter*	**162″**
Shelf surface	**109 sq ft**	*Running shelf*	**106 ft**
Drawer bases	**132″**	*Blind corner*	**30″**

What Works

- Full 162″ counter with a 39″ aisle, above the 36″ walkway minimum.
- 8″ shelves hold cans, jars, and spices one or two deep, so nothing hides behind anything.
- Drawers open with room to stand.

What to Watch

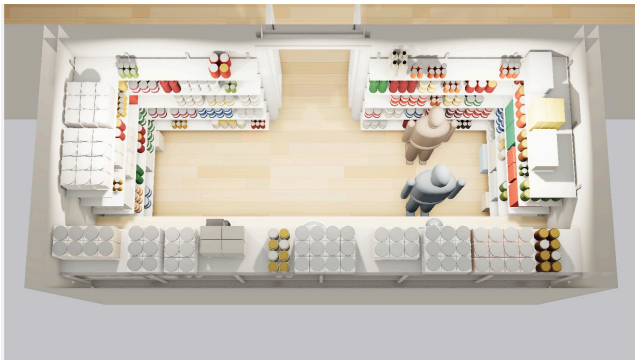
- Less shelf area than A or B.
- Cereal boxes and bins do not fit the 8″ shelves; they go on the end walls.
- Passing someone is still a squeeze.



From the doorway



At the workstation

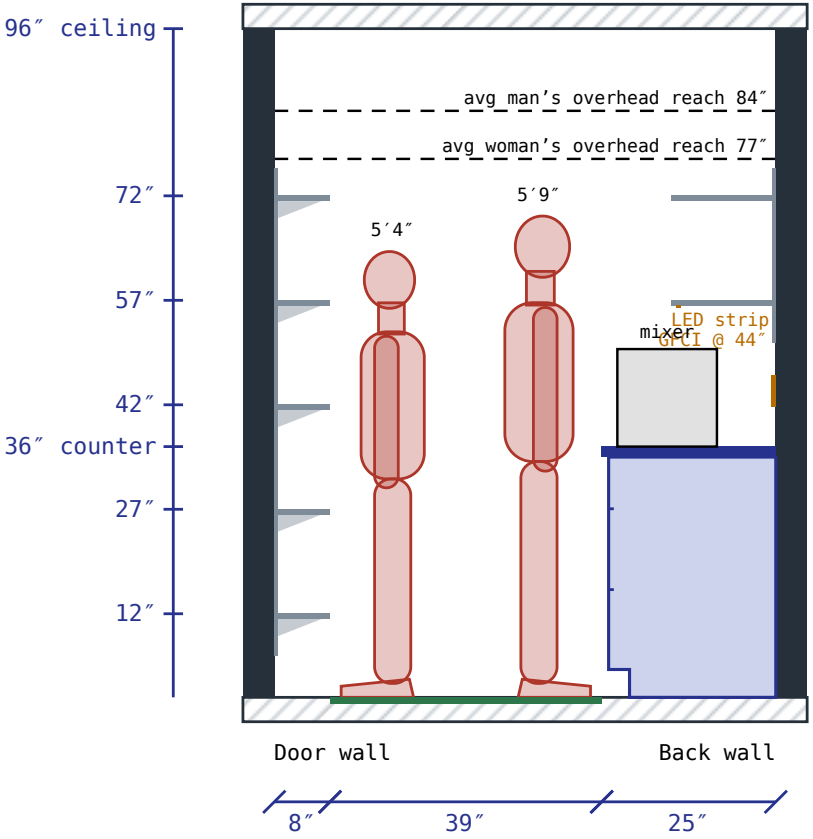


Overhead cutaway, kitchen at top

Layout D: Sections

Cut through the room and viewed from the side. Heights are inches above the floor; the colored floor bar marks the clear aisle.

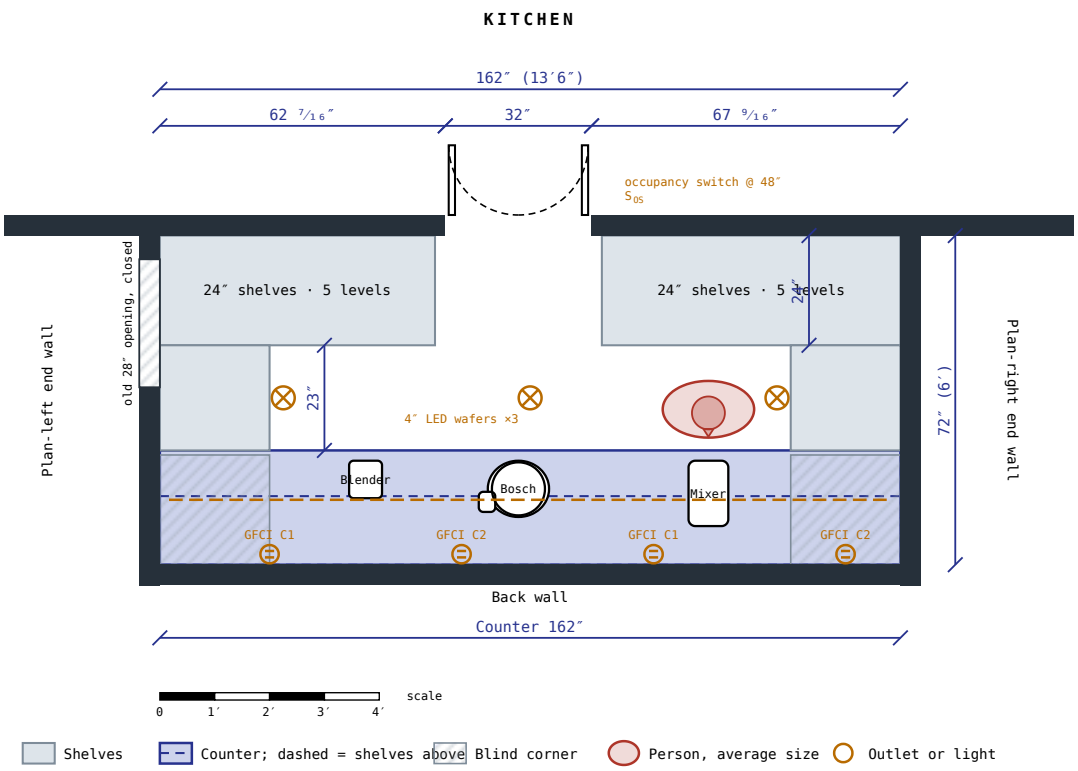
Cross-Section: 39″ Aisle With the Can Wall



Layout E: Deep Everything

Not realistic

24″ shelves on the door wall plus a 25″ counter on the back wall. It looks like more storage on paper, but it leaves a 23″ slot to stand in.



Narrowest aisle	**23″**	*Counter*	**162″**
Shelf surface	**177 sq ft**	*Running shelf*	**98 ft**
Drawer bases	**114″**	*Blind corner*	**48″**

What Works

- Most raw cubic storage.

What to Watch

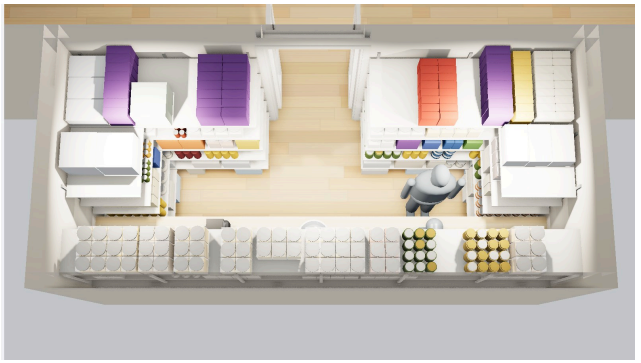
- 23″ aisle: an average man's shoulders (20″) clear it by 1½″ a side.
- You cannot crouch to a bottom shelf or open a 22″ drawer and stand in front of it.
- Items 24″ deep get buried at the back.



From the doorway



At the workstation

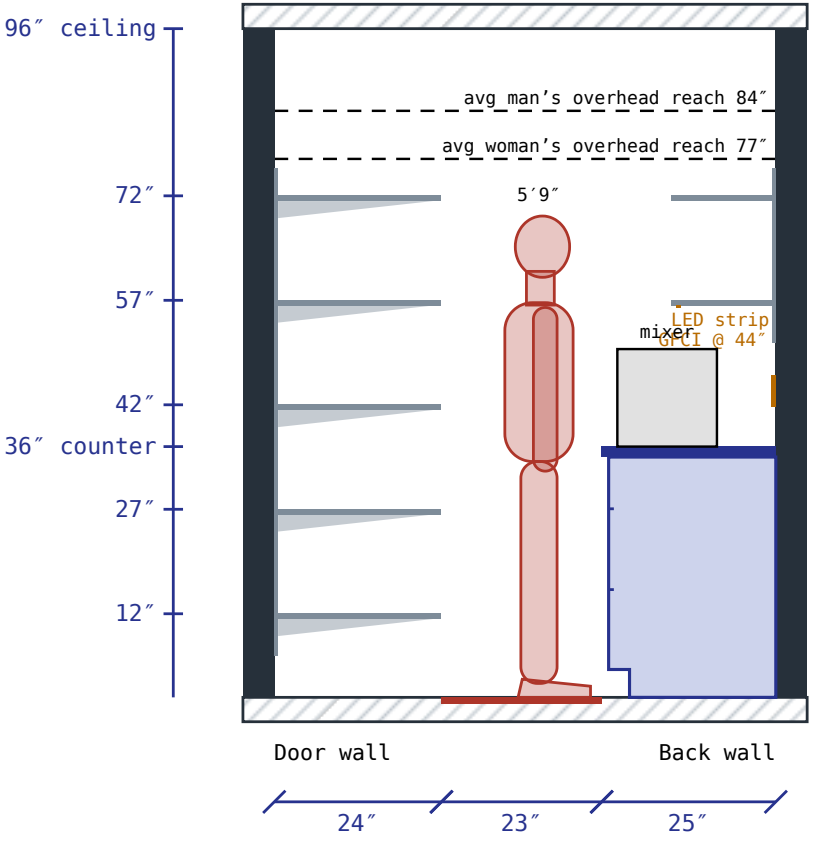


Overhead cutaway, kitchen at top

Layout E: Sections

Cut through the room and viewed from the side. Heights are inches above the floor; the colored floor bar marks the clear aisle.

Cross-Section: 23″ Aisle



Electrical Plan

Confirm against the code edition on the permit. New Jersey adopted UCC 2024 in August 2026; its rule proposal moves the electrical subcode to the 2023 NEC, and work already under permit can generally finish under the 2020 NEC.

1. *Two counter receptacles, mounted at about 44″.* That puts them 8″ above the 36″ counter and well under the first shelf at 57″. In a pantry, code requires countertop receptacles placed so no point along the wall is more than 24″ from one, no more than 20″ above the counter, and fed by 20 A small-appliance circuits. *NEC 210.52(B), (C)(1), (C)(5)*
2. *Split them across two 20 A circuits, or give the counter its own.* The Bosch at 500 W draws about 4 A. A full-size blender can pull 10 to 12 A. Share one circuit with a kitchen toaster and the breaker will trip mid-knead. *Bosch spec sheet; 500 W ÷ 120 V*
3. *GFCI-protect the counter receptacles.* Code requires it for receptacles serving countertops. Use a GFCI breaker or put the GFCI device where its reset button stays reachable, not behind the mixer. *NKBA Guideline 30*
4. *Set receptacles between appliance spots, not behind them.* The Bosch is 12 $\frac{5}{16}$″ deep with its drive tower at the right rear. The plan's receptacle positions leave every plug visible. *Plan, electrical layer*
5. *Light the aisle, not the shelf tops.* Three 4″ LED wafers on the aisle centerline, 54″ apart, light the shelf faces. Choose 90+ CRI so labels read true. *Plan, electrical layer*
6. *Add an LED strip under the first shelf over the counter.* The counter otherwise sits in your own shadow. NKBA calls for task lighting at every work surface. *NKBA Guideline 31*
7. *Use an occupancy switch on the kitchen side of the door.* Inside, shelving fills the wall beside the casing, and your hands are usually full. A sensor switch at 48″ turns the lights on as you walk in. *Layout geometry*
8. *Rough in while the walls are open.* In Layout A the counter wall is the closed-up 28″ opening, so the electrician wires new framing. Have the framer add 2× blocking at shelf-standard locations on every shelf wall; a stocked 15″ shelf is heavy. *Layout A*

Assumptions to Confirm

- *Room 162″ × 72″ inside.* Read from your plan's 162″ and 72″ interior dimensions; the 162″ bottom string matches the top.
- *Door location.* The opening is 32″ wide, starting 62 $\frac{7}{16}$″ from the plan-left wall with 67 $\frac{9}{16}$″ beyond it. The fractions on the drawing are small; these are the values that sum to 162″. Confirm on site.
- *French doors swing into the kitchen.* With two 15¼″ leaves inside a 32″ opening, about 27¾″ of clear passage remains with both open, enough for 20″ shoulders. If the doors swung into the pantry, each leaf would block roughly 16″ of door-wall shelving.
- *The old 28″ opening is closed flush* and finished like the rest of the wall.
- *Flat 96″ ceiling* with no soffits, ducts, or access panels in the room.
- *Shelves are ¾″ boards on wall standards* (adjustable), set back 2¼″ from the door opening to clear the casing. Levels are the top surface of each shelf. Supports sit no more than 30″ apart (a common rule of thumb against sag).
- *Counter is 36″ high and 25″ deep* (24″ drawer bases plus a 1″ overhang). The "Deep shelf" toggle swaps the cabinets for a 24″-deep work shelf at 36″ on heavy brackets.
- *Appliances:* Bosch Universal Plus at true size. The blender (≈7¼″ × 8¼″ × 16½″) and second mixer (≈8¾″ × 14¼″ × 14″, tilt-head) are typical-size stand-ins.



Bosch Universal Plus at true size: 15⅜″ W × 12 $\frac{5}{16}$″ D base, 11″ tall with bowl and lid.